Feature taxonomy for addition at "="

How to read an operand plot

Each plot is one feature at the "=" token of the prompt "calc: a+b=". Vertical axis: a (first operand). Horizontal axis: b (second operand). Both run 0 to 99, so the plot covers all 10,000 prompts. Color: how strongly the feature fires on that prompt. White: it does not fire.

Directions in the plot

Top-left to bottom-right: a + b is constant (sum). Bottom-left to top-right: a - b is constant (the a = b diagonal is a - b = 0). Vertical lines: b is constant. Horizontal lines: a is constant.

Two labels per feature

Role (Anthropic's kinds): how the feature combines a and b, read from the shape. Family: what kind of condition it checks (last digit, magnitude, mod 100, comparison).

Step 1, the role: ask in order, stop at the first yes

1. Almost the whole plot filled (more than 90%)? -> mostly active. 2. Diagonal lines or bands, top-left to bottom-right? -> sum. 3. Only vertical bars, or only horizontal bars? -> add function. 4. L-shapes, corners, nested squares, or the a = b diagonal? -> comparison. 5. A regular grid, or a few separate blobs? -> lookup table. 6. None, several mixed, or a shape only in part of the plot? -> unclear.

Step 2, the family

last digit: the pattern repeats every 10. magnitude: one or a few smooth regions. mod 100: diagonal lines exactly 100 apart, or one thin diagonal line. comparison: about which operand is larger, or their difference. unclear: none of these.

How the code decides (for reference)

For each candidate variable v(a, b) the code computes R^2: the share of the feature's variation explained by the average activation within each value of v. It does this on the activation values and on the on/off pattern, and keeps the better one. A feature gets the category of its best variable only if R^2 >= 0.6 and it leads every other category by >= 0.15. A variable that contains a simpler one (e.g. (a+b) mod 100 contains (a+b) mod 10) must beat it by 0.05, and a joint lookup must beat the additive model (effect of a + effect of b) by 0.1. Otherwise: mixed or unclear.

When unsure

Write unclear and say why in the note column. A careful "unclear" is better than a guess.

add function / magnitude

Math

```
Anthropic: unions of horizontal lines (a condition on a) and vertical lines (a
condition on b).
Typical: 1[a mod 10 = 9  OR  b mod 10 = 9]  ("add 9"), scored here by the
additive model
g1(a mod 10) + g2(b mod 10).  Ours: 1[b in [lo, lo+L)] or floor(b/10) (or the
same for a).
```

What it means

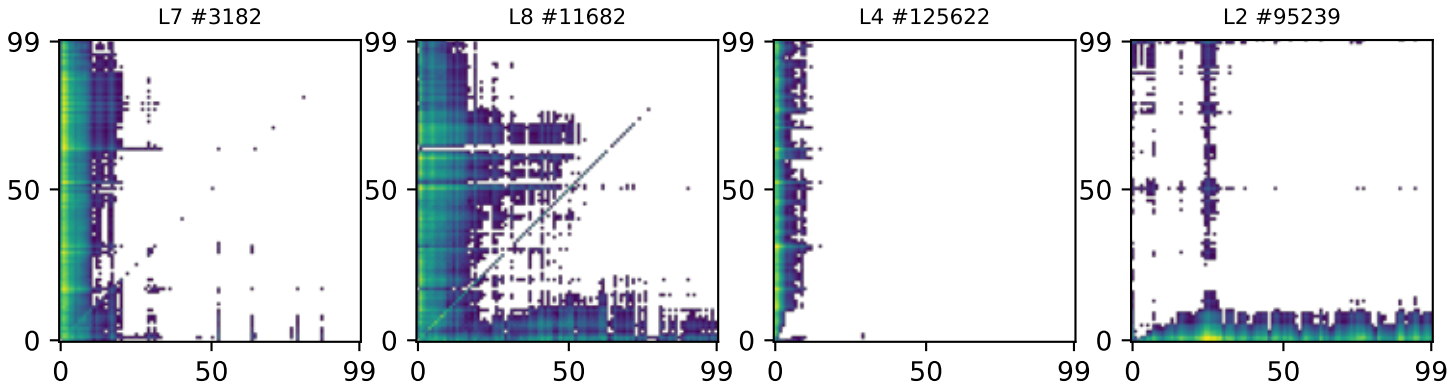
Reads one operand on its own, before the two are combined. Anthropic's typical example, "add 9", fires whenever either operand ends in 9 (bottom left: a synthetic plot of that pattern). At "=" in Llama we found NONE of that kind: the method can detect it (a synthetic "add 9" scores 0.95 on the additive model), but no real feature does. The add function features we did find are the other kind: "b is a single digit" (b < 10), a single thick bar (examples below). The closest relative of "add 9" in Llama may be the nested squares (comparison / last digit): the same lines, cut off at the a = b diagonal.

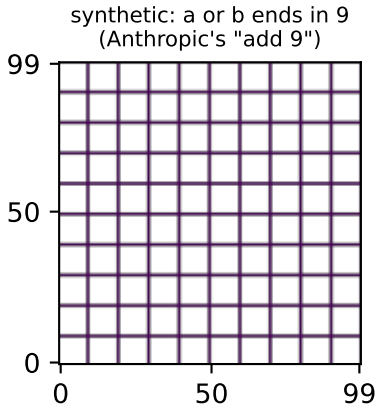
How it looks

Anthropic's kind: thin full-length lines every 10, in both directions (bottom left). Llama's kind: one vertical bar at the left edge (b small), or one horizontal bar at the bottom (a small).

Confused with

Nested squares (comparison / last digit): lines every 10 that stop at the a = b diagonal. Full-length lines -> add function; lines that turn a corner at a = b -> comparison.





Bottom left: Anthropic's pattern, drawn from the math (not a Llama feature).
Top row: the add function features we found in Llama.

a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

sum / last digit

Math

`v(a, b) = (a + b) mod 10`

What it means

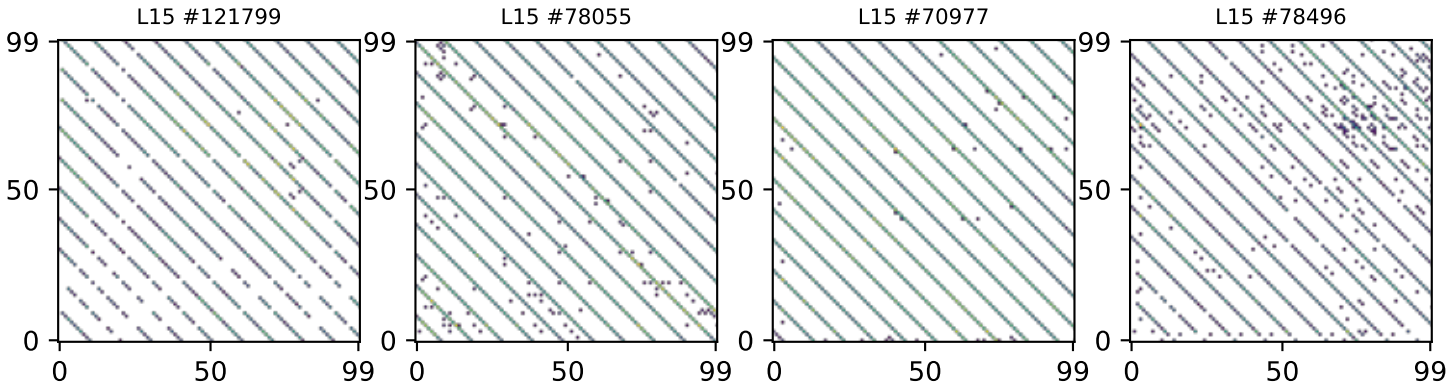
The units digit of the answer: 36 + 59 = 95 -> 5. This is the output side of the last-digit pathway.

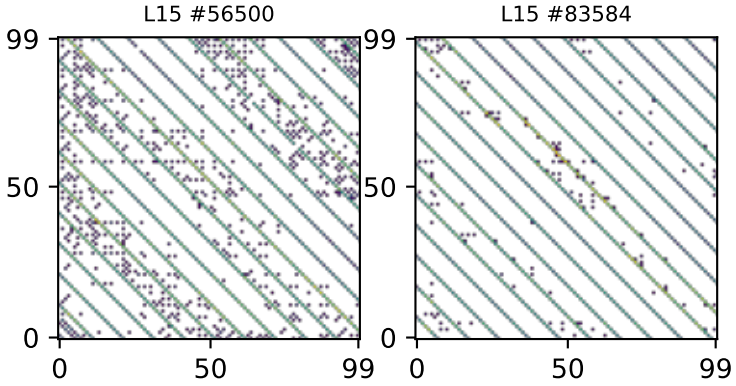
How it looks

Thin diagonal lines (top-left to bottom-right), one every 10 steps of a + b.

Confused with

mod 100 features: count the gap between lines. Every 10 -> last digit; 100 apart -> mod 100.





a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

sum / mod 100

Math

`v(a, b) = (a + b) mod 100`

What it means

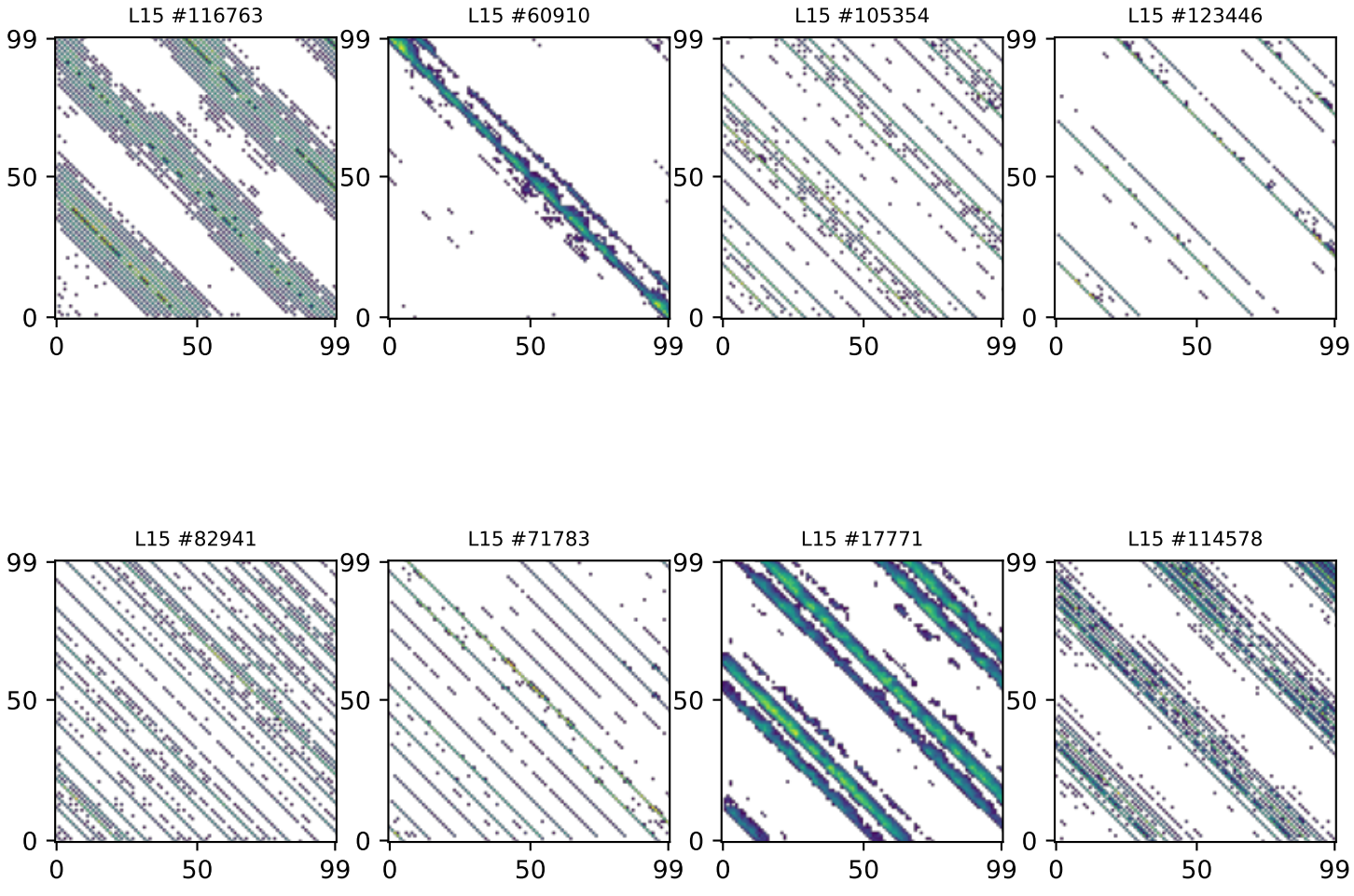
The last two digits of the answer: 95 and 195 are the same value (95). Fits the period-100 components found in the number embeddings.

How it looks

Diagonal lines or bands exactly 100 apart (e.g. a + b = 55 and 155), or one thin line (an exact sum below 100, e.g. a + b = 99).

Confused with

sum / magnitude: a single wide band is a range of sums, not mod 100.



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

sum / magnitude

Math

`v(a, b) = 1[a + b in [lo, lo+L)]  or  floor((a + b)/10)`

What it means

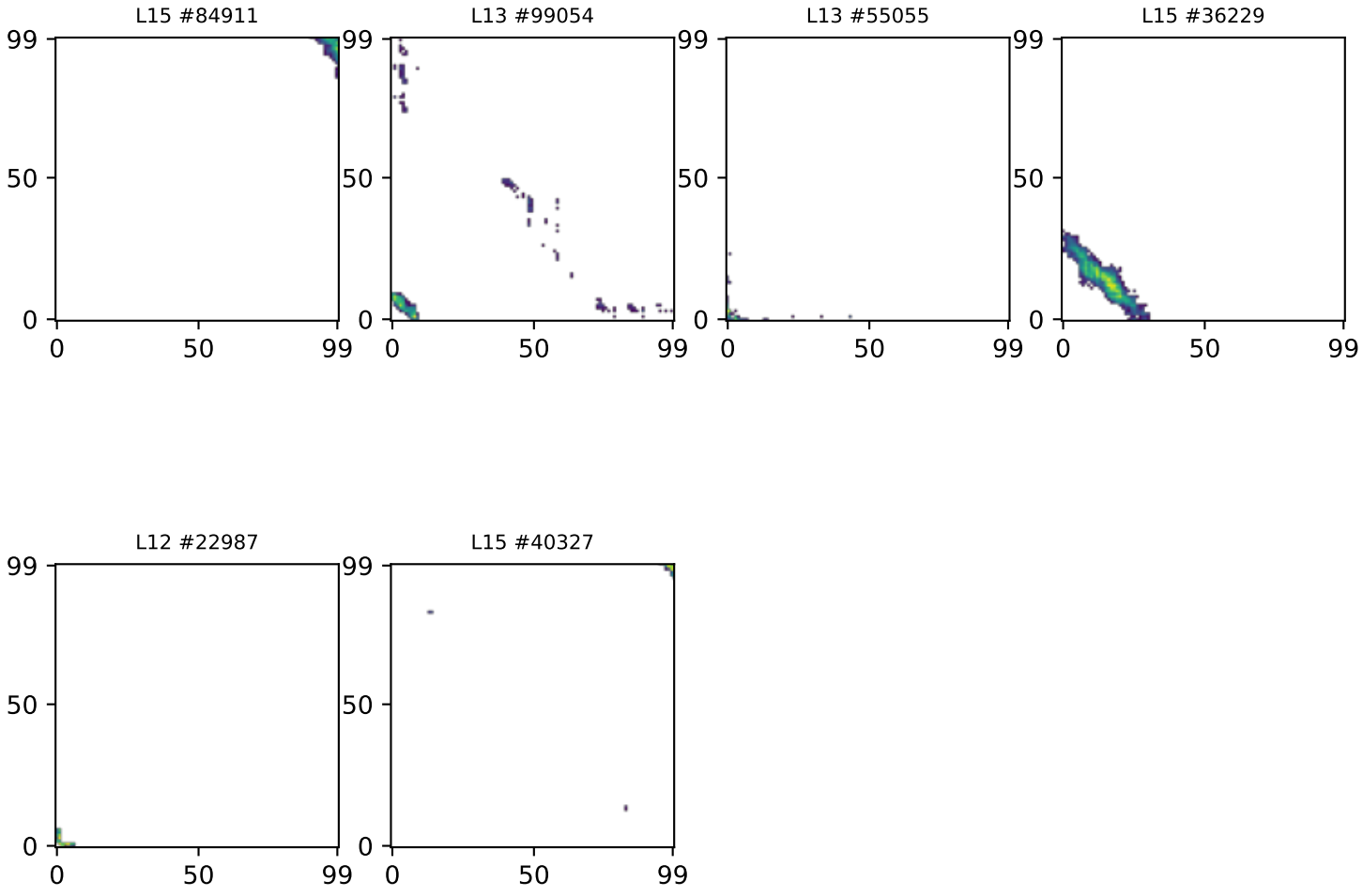
The rough size of the answer: "the sum is about 30". The low-precision side of addition.

How it looks

One wide diagonal band, or a corner (bottom-left: small sums; top-right: sums near 198).

Confused with

mod 100 (several lines 100 apart) and mostly active (a band that fills over half the plot).



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

lookup table / last digit

Math

`v(a, b) = (a mod 10, b mod 10) jointly: 100 combinations (e.g. "_6 + _9");`
`also carry = 1[(a mod 10) + (b mod 10) >= 10].`

What it means

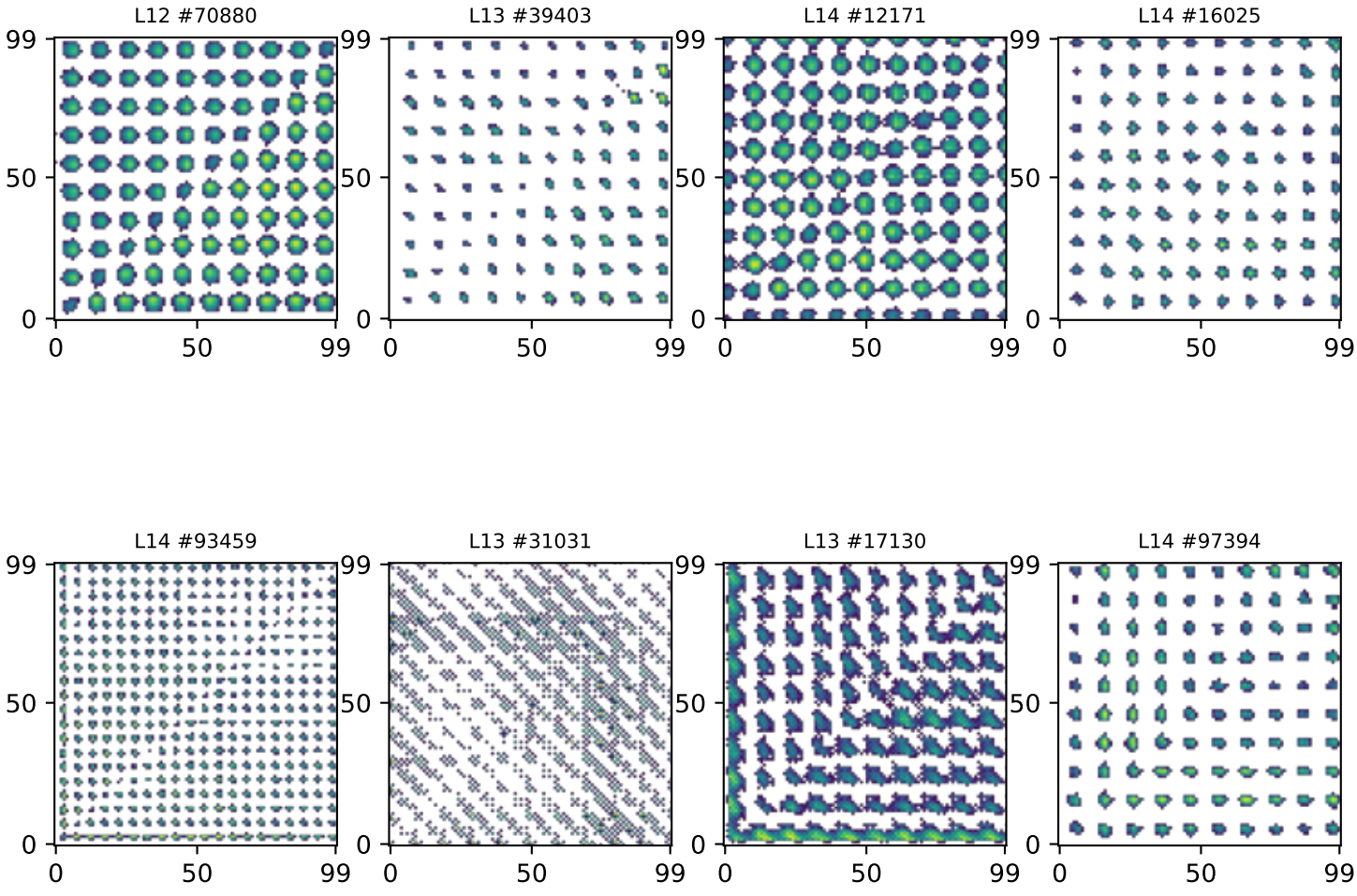
Combines the two units digits before the sum is formed, like a times table for units. Anthropic's "_6 + _9" features. Part of the last-digit pathway.

How it looks

A regular grid of dots or blobs repeating every 10 in both directions.

Confused with

Dense dot textures that fill the plot (mostly active), and nested squares (comparison / last digit).



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

lookup table / magnitude

Math

`v(a, b) = (floor(a/10), floor(b/10)) jointly, and it must clearly beat the additive model`
`g1(floor(a/10)) + g2(floor(b/10)): a real interaction between a and b.`

What it means

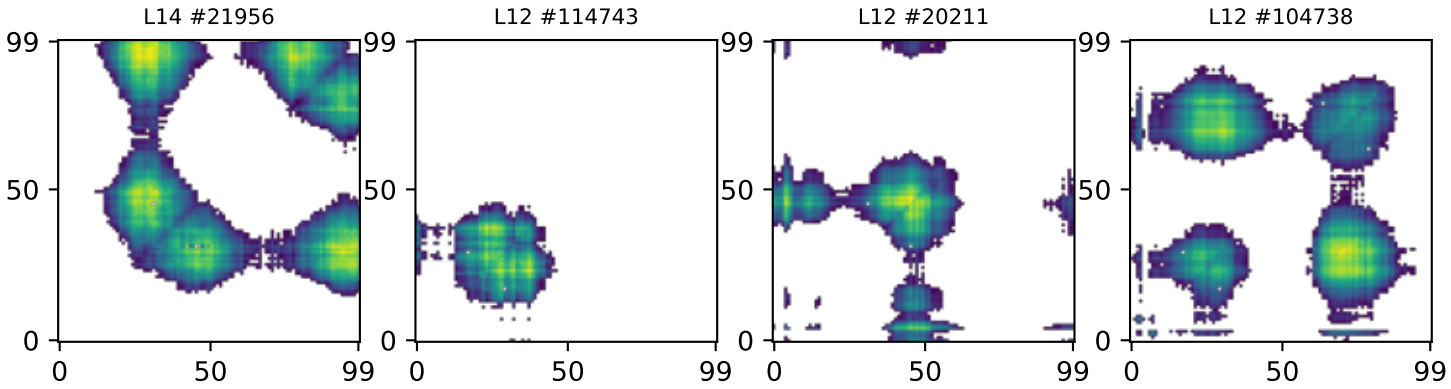
"a is about X AND b is about Y": a low-precision lookup (Anthropic's "~36 + ~60").

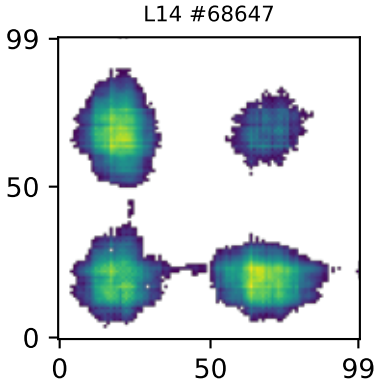
How it looks

One or a few separate, compact blobs at specific (a, b) locations. Mirror pairs, e.g. (a about 75, b about 30) and (a about 30, b about 75), are normal: addition is commutative. All lookups of this kind were checked by hand (results/manual_review/).

Confused with

L-shapes whose arms meet on the a = b diagonal, even when drawn as beads (comparison); bands from top-left to bottom-right (sum); bars crossing the whole plot (add function). Test: can it be described as "a about X and b about Y", and nothing else?





a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

comparison / comparison

Math

```
v(a, b) = min(a, b) or max(a, b) in a range or in decades;  a - b in a range;
1[a < b];  1[a = b]
```

What it means

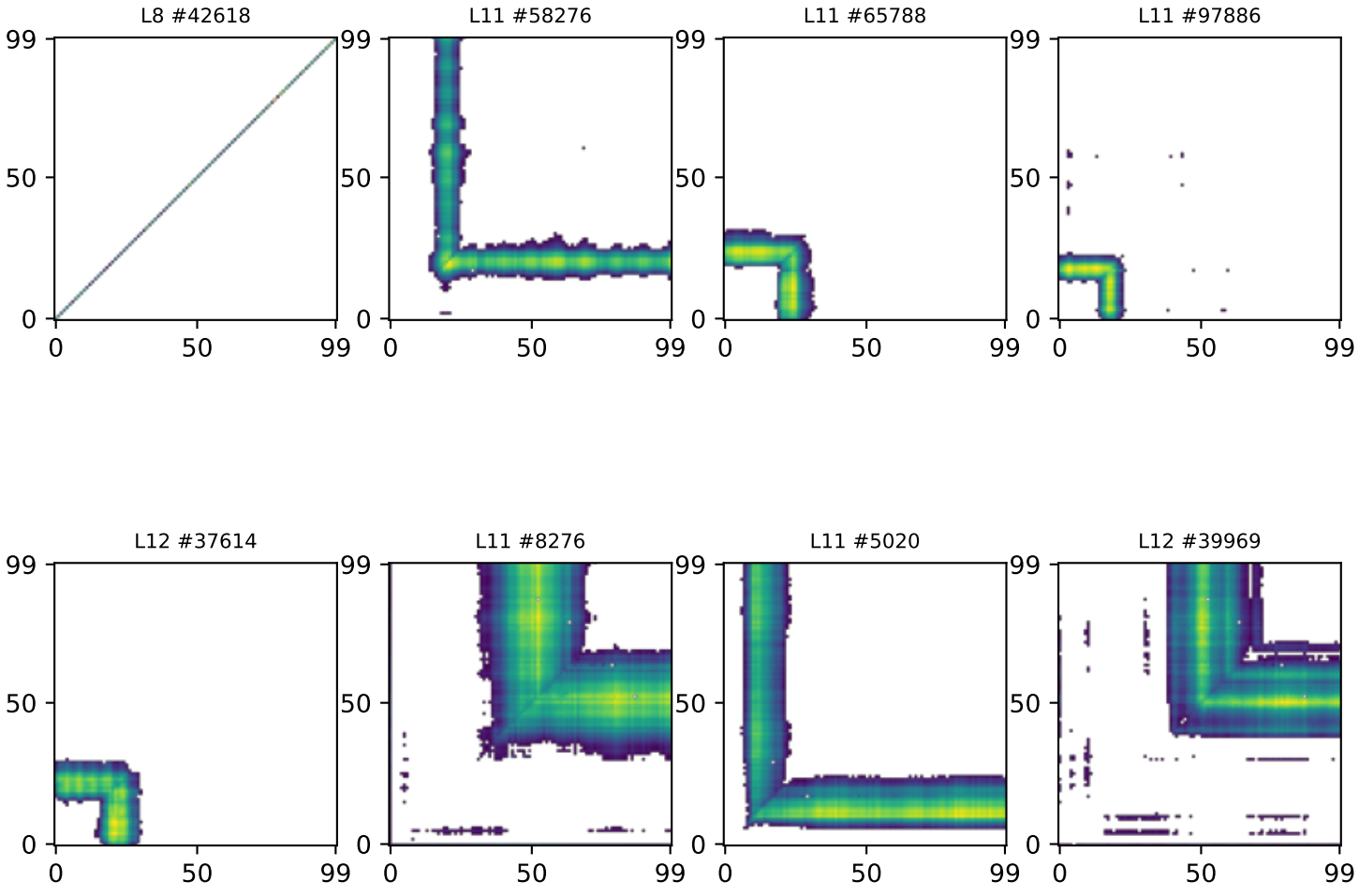
Which operand is larger, or how far apart they are: "the smaller operand is about 20".

How it looks

L-shapes or corners that bend exactly at the a = b diagonal; the a = b diagonal itself (bottom-left to top-right); a region above or below that diagonal.

Confused with

Blobs (lookup table). An L-shape whose corner lies on a = b is comparison.



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

comparison / last digit

Math

`v(a, b) = max(a, b) mod 10  or  min(a, b) mod 10`

What it means

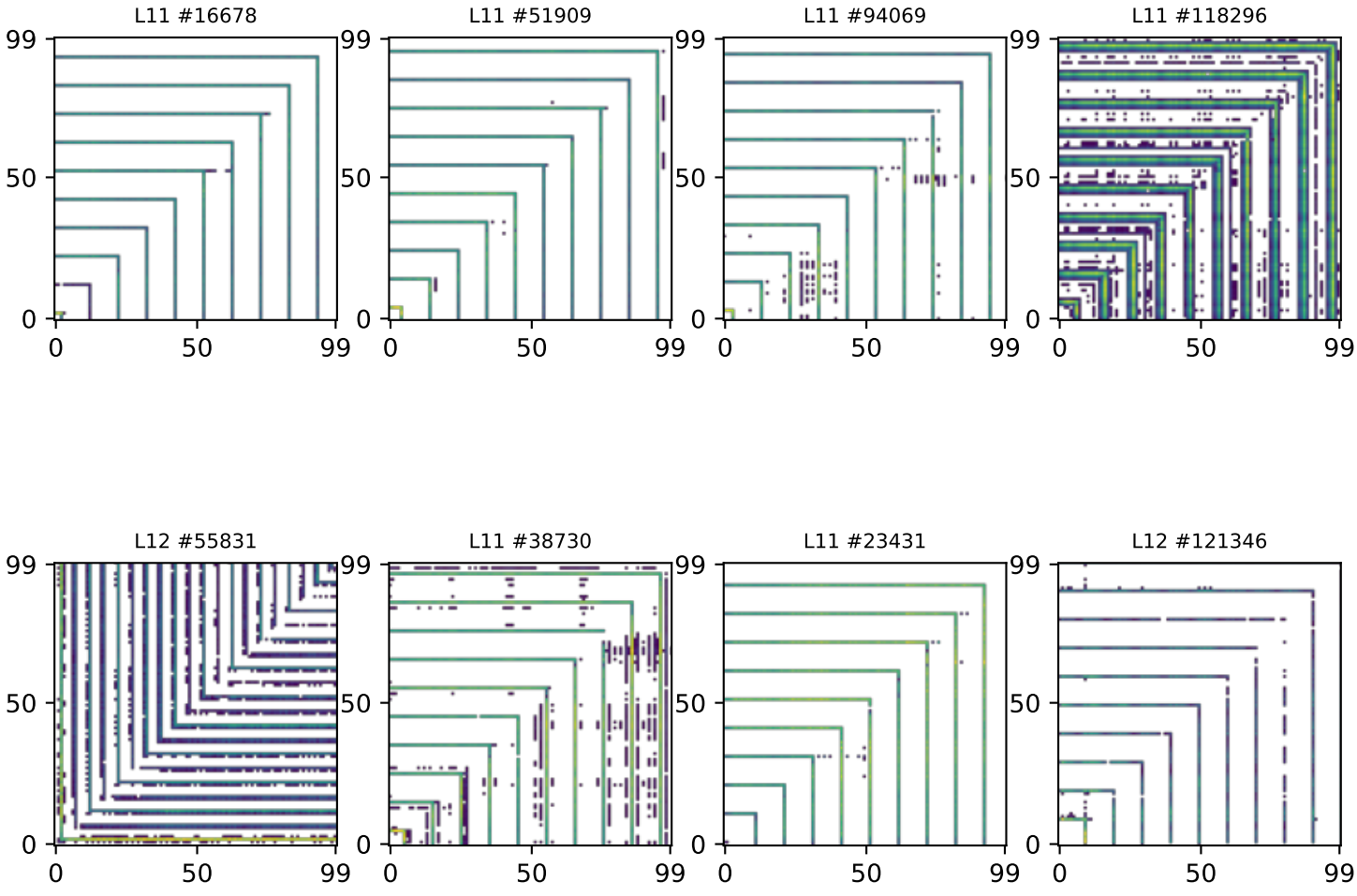
The last digit of the larger (or smaller) operand. The pilot found "max(a, b) ends in d". "max(a, b) ends in 9" = "a ends in 9 and a is larger, or b ends in 9 and b is larger": Anthropic's "add 9" lines cut off at the a = b diagonal (hypothesis: Llama's version of add function).

How it looks

Nested squares: lines that turn a corner at the a = b diagonal, one every 10.

Confused with

Lookup grids (dots in rows and columns, no corners), and Anthropic-style add function features (lines every 10 that run the full length without turning at a = b).



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

mostly active

Math

```
n_prompts / 9,992 > 0.9   (Anthropic: "active on the '=' token of most of our
10,000 prompts";
we use 90%: at 50%, the first blind check showed clear shapes such as a < b
being hidden)
```

What it means

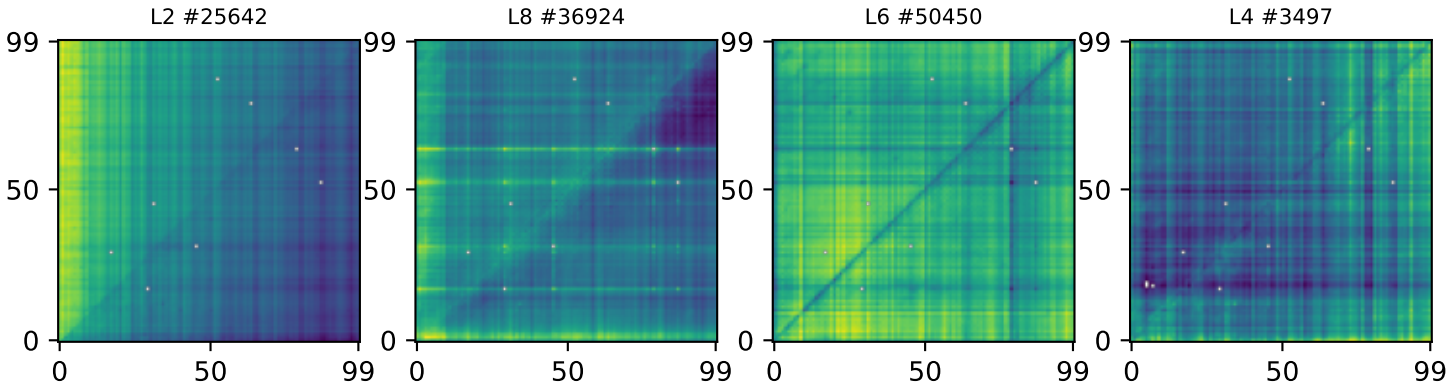
A feature that is on for most addition prompts. Its value can still vary a little with a or b, but its shape says nothing, so it gets this role whatever it looks like.

How it looks

Almost the whole plot filled, often with a texture or a faint a = b diagonal.

Confused with

A plot filled on about half (e.g. one side of the a = b diagonal) is NOT mostly active: label its shape (that one is comparison).



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.

unclear

Math

```
No variable reaches R^2 >= 0.6, or two categories score about the same
("mixed").
```

What it means

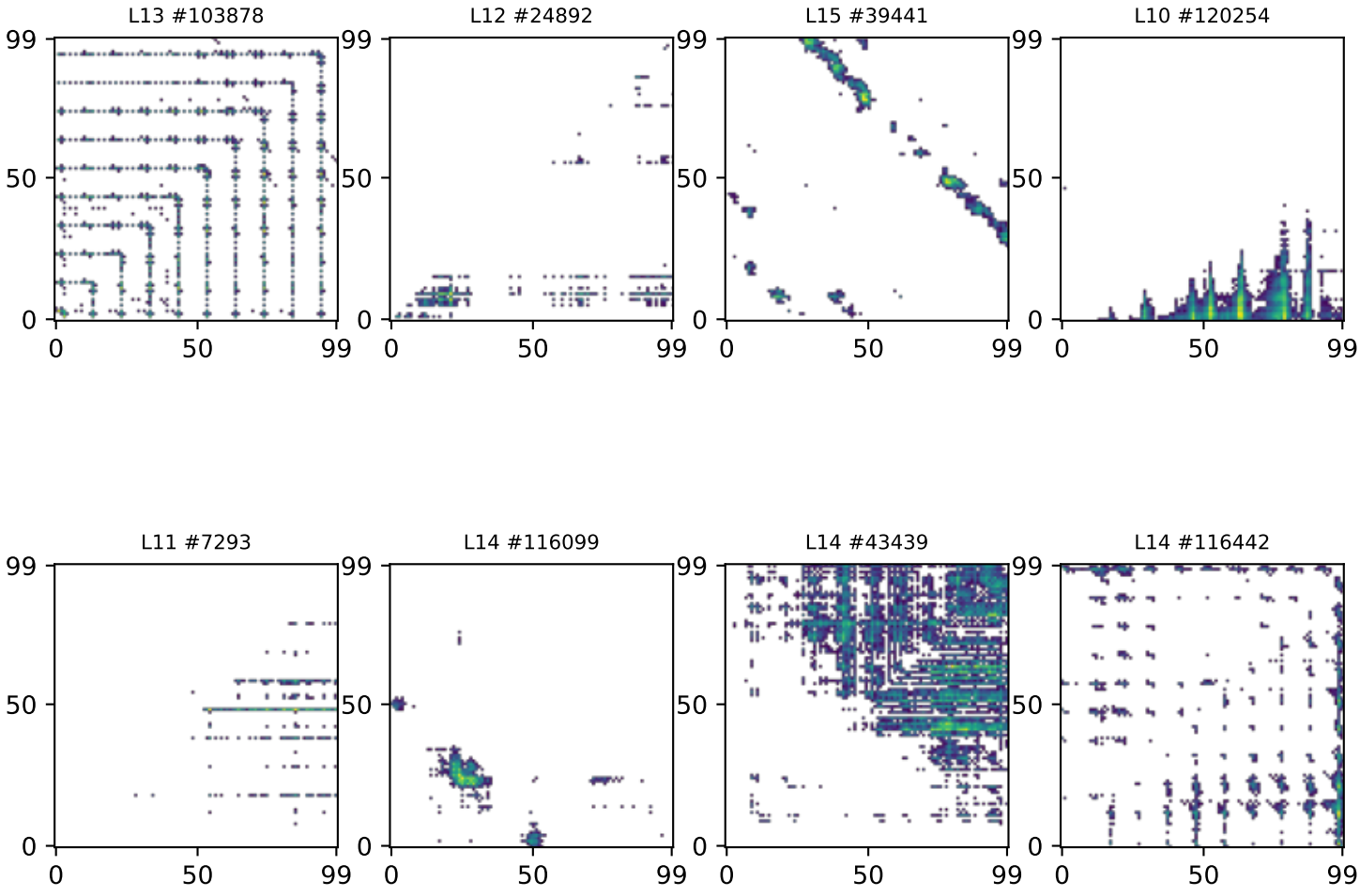
Either noise (scattered specks), two conditions at once (a lattice only in part of the grid), or a kind the variable library does not cover.

How it looks

Specks, several shapes mixed, or a clear shape that only covers part of the plot.

Confused with

Nothing: when in doubt, this is the right answer.



a on the vertical axis, b on the horizontal axis, white = not active. Examples are outside the blind set.